

# Scale recipes

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: A recipe for 6 uses 450 g flour. How much flour is needed for 14? Copy or complete the first step.

2. In the ratio 4:7, the smaller part is 20. What is the larger part?

3. Share £420 in the ratio 3:4.

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: A recipe for 6 uses 450 g flour. How much flour is needed for 14? Copy or complete the first step.  
**Answer** 1,050 g.
2. In the ratio 4:7, the smaller part is 20. What is the larger part?  
**Answer** 35.
3. Share £420 in the ratio 3:4.  
**Answer** £180 and £240.
4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.  
**Answer** Apply the same multiplicative relationship to every corresponding measure or ratio part.

# Scale recipes

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. A recipe for 6 uses 450 g flour. How much flour is needed for 14?

6. Check this result using an inverse, estimate or known fact: Share £420 in the ratio 3:4.

2. In the ratio 4:7, the smaller part is 20. What is the larger part?

7. Prove or explain your reasoning: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

3. Share £420 in the ratio 3:4.

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.

4. Solve this independently and show every stage: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: In the ratio 4:7, the smaller part is 20. What is the larger part?

10. Write and solve a similar question to: Share £420 in the ratio 3:4.

# Teacher answers and checking prompts

1. A recipe for 6 uses 450 g flour. How much flour is needed for 14?  
**Answer** 1,050 g.
2. In the ratio 4:7, the smaller part is 20. What is the larger part?  
**Answer** 35.
3. Share £420 in the ratio 3:4.  
**Answer** £180 and £240.
4. Solve this independently and show every stage: A recipe for 6 uses 450 g flour. How much flour is needed for 14?  
**Answer** 1,050 g.
5. Use a second method or representation for: In the ratio 4:7, the smaller part is 20. What is the larger part?  
**Answer** Expected result: 35. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Share £420 in the ratio 3:4.  
**Answer** Expected result: £180 and £240. A valid check must be shown.
7. Prove or explain your reasoning: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.  
**Answer** 52 km. The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.  
**Answer** Apply the same multiplicative relationship to every corresponding measure or ratio part.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Share £420 in the ratio 3:4.  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Scale recipes

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Prove or explain your reasoning: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

6. Create a new example that tests the same idea as: Share £420 in the ratio 3:4.

2. Prove the answer to this question, rather than only calculating it: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: In the ratio 4:7, the smaller part is 20. What is the larger part?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: £180 and £240.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Prove or explain your reasoning: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.  
**Answer** 52 km. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A recipe for 6 uses 450 g flour. How much flour is needed for 14?  
**Answer** Expected result: 1,050 g. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: In the ratio 4:7, the smaller part is 20. What is the larger part?  
**Answer** Expected result: 35. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: £180 and £240.  
**Answer** One valid reconstruction is: Share £420 in the ratio 3:4. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.  
**Answer** Apply the same multiplicative relationship to every corresponding measure or ratio part.
6. Create a new example that tests the same idea as: Share £420 in the ratio 3:4.  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.